Stuck Pipe Prevention: Early Warning Signs and First Response

Lab sample document (synthetic). Not operational guidance.

Purpose

This guide describes the surface indicators that commonly precede a stuck pipe event and the first actions a driller should consider. It accompanies the HOL_PTEN_ML stuck-pipe risk model, which scores each stand on hookload, torque, rate of penetration (ROP) and rotary speed (RPM) trends.

Mechanisms

Differential sticking occurs when the drill string rests against a permeable formation and the overbalance pressure holds it in the filter cake. It is most likely while the string is stationary, for example during connections or surveys.

Mechanical sticking includes pack-off from poor hole cleaning, wellbore geometry problems such as key seats and ledges, and collapse of unstable shale.

Early warning signs

Watch for these trends over the last two to three stands:

- **Torque rising** while drilling parameters are unchanged. A sustained rise of 15 to 25 percent over baseline torque is a common precursor of pack-off.
- **Overpull on connections.** Hookload above the expected pick-up weight by more than 20 to 30 klb indicates drag or a developing restriction.
- **ROP falling** without a formation change, often together with erratic torque.
- **Standpipe pressure (SPP) increasing** at constant flow, which suggests the annulus is loading with cuttings.
- **Reduced cuttings returns** at the shakers relative to ROP.

A single indicator is rarely conclusive. Two or more indicators trending together within the same stand should prompt action.

First response checklist

1. Stop drilling ahead and keep the string moving. Do not leave pipe stationary.
2. Circulate at full rate and rotate if hole conditions allow, to clean the annulus.
3. Work the pipe within safe limits; apply torque before jarring if mechanically stuck.
4. Record hookload, torque and SPP at each action so the event can be reviewed.
5. Notify the drilling supervisor and the remote operations center.

Connection practice in high-risk intervals

- Minimize time on slips; target connection time under 5 minutes.
- Back-ream or wash the last stand before the connection when torque has been trending upward.
- Confirm pick-up and slack-off weights against the torque and drag model at every connection.